Booklet No.

Booklet No.

1.2) $\delta(q_1, \epsilon, A) = \{(q_1, aB), (q_1, \epsilon$

$\delta(q_1, \epsilon, B) = (q_1, Aa)$

1.3) $S \to aSbb / A / B$, $A \to aA / a$, $B \to bB / b$

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Go, change the world

Academic year 2024-2025 (Odd Sem)
DEPARTMENT OF

Computer Science & Engineering

Date	8 Jan 2025	Maximum Marks	10+50
Course Code	CS354TA	Duration	120 Min
Sem-V	CIE-II	Staff: HKK|ASP|SMS|MRA	

THEORY OF COMPUTATION
(Common to CSE & ISE)

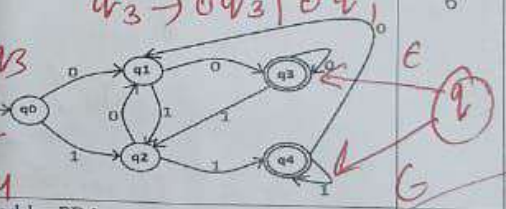
PART-A

$S \to (L) / a$, $L' \to , SL' / \epsilon$, $L = SL'$

		Marks	BT	CO
1.1	Define left recursion. Eliminate left recursion from the following grammar S→(L) | a, L→L,S | S	2	L2	CO4
1.2	Convert the given CFG to PDA S→a |aA | B, A→aB | ε, B→Aa. $\delta(q_0, \epsilon, Z) = (q_1, SZ)$ $\delta(q_1, \epsilon, S) = \{(q_1, a), (q_1, aA), (q_1, B)$	2	L2	CO3
1.3	Find context-free grammars for the languages L={(a^n b^m : n ≠ 2m}.	2	L3	CO2
1.4	Consider the following grammar with S as the start variable and a, ; ,< , > are as terminals. S→<L|a, L→aR|<LR, R→>| ; L. How many distinct parse trees are there for the string <<a>; a>?	1	L3	CO2
1.5	Let G be the CFG and let "L" denotes the number of left most derivations, "R" denotes the number of right most derivations and "P" denotes the number of parse trees. Assume L, R, P are computed for a particular string. For a given CFG G and a string w, what is the relation between L, R, P (use relational operators)? $L = R - P$	1	L1	CO1
1.6	Define DPDA.	2	L1	CO1

PART-B

2.a	Construct PDA that accepts the language L = {Na(w)=Nb(w)+ Nc(w)} on Σ={a,b,c}. Give the graphical representation for PDA obtained. Show the moves made by the PDA for the string caacaabc.	7	L3	CO4
2.b	Construct a finite automaton that accepts the language generated by the right linear grammar S -> cA | baS, A -> bB | aC | ε, B -> aA |bbC |ε , C-> abA | baC | aS	3	L2	CO1
3.a	Find a left-linear grammar for the language accepted by the finite automata below	6	L3	CO3
3.b	Write a note on the languages accepted by PDA	4	L1	CO2
4.a	Define CNF. Transform the grammar with productions $S \to abAB$, $A \to bAB \| \lambda$, $B \to BAa \| A \| \lambda$ into Chomsky normal form	6	L3	CO3
4.b	Define ambiguous grammar. Show that the following grammar is ambiguous and Construct an unambiguous grammar equivalent to this grammar. S→AB | aaB, A→a | Aa, B→b.	4	L4	CO4



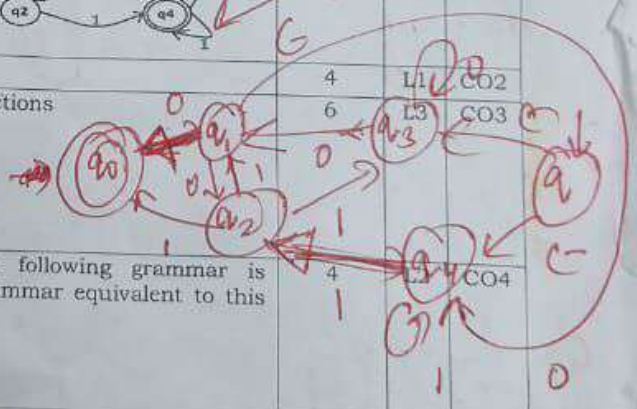
$q_3 \to 0q_3 | 0q_1$

$q_0 \to \epsilon$

$q_2 \to 0q_1 | 0q_3$

$q_1 \to 0q_0 | 0q_4 | 0q_2 | 0$ $q_4 \to 1q_4 | 1q_2$

$q_2 \to 1q_0 | 1q_3 | 1q_1 | q_5 \to q_3 | q_4$



1.3 {a, any no of a, b, any no b, aabb, abbb, a---abb, abbb--}

1.2) $\delta(q_0, \epsilon, Z_0) = (q_1, SZ_0)$, $\delta(q_1, a, S) = (q_1, \epsilon)$, $\delta(q_1, a, S) = (q_1$

$\delta(q_1, a, A) = (q_1, B)$, $\delta(q_1, \epsilon, A) =$ $\delta(q_1, a, B) = (q_1$

push: $\delta(q_i, a, Z) = (q_j, AB)$
$(q_i Z q_k) \to a(q_j A q_0)(q_0 B q_k)$

Pop: $\delta(q_i, a, Z) = (q_j, \epsilon)$
$(q_i Z q_j) \to a$

5	Give an algorithm to convert PDA to CFG. Convert the following Pushdown Automata transitions to Context Free Grammar. Describe the language accepted by the CFG. 1. δ (q0,0,Z)=(q0,AZ) 2. δ (q0,1,Z)=(q0,BZ) δ (q0,0,B)=(q0, ε) δ (q0,1,A)=(q0, ε) 3. δ (q0,0,A)=(q0,AA) 4. δ (q0,1,B)=(q0,BB) δ (q0, ε,Z)=(q1, ε)	10	L4	CO3
6.a	Let L1= {a^i b^j c^k | i=j} and L2={ a^i b^j c^k | j=k }. Find L3=L1UL2 and L4=L1∩L2 . Show that L1, L2 and L3 are CFLs but L4 is not CFL.	6	L2	CO2
6.b	Discuss the applications of Context free grammars.	4	L2	CO1

count = no of 0 = 2
no of 1 = 3

$q_0 Z q_0 \to 0$
$q_0 A q_0 \to 1$
$q_0 Z q_1 \to \epsilon$

1. $q_0 Z q_0 \to 0[q_0 A q_0][q_0 Z q_0] \mid 0[q_0 A q_1][q_1 Z q_0]$
$q_0 Z q_1 \to 0[q_0 A q_1][q_1 Z q_1]$
2. $q_0 Z q_0 \to 1[q_0 B q_0][q_0 Z q_0] \mid 1[q_0 B q_1][q_1 Z q_0]$
$q_0 Z q_1 \to 1[q_0 B q_1][q_1 Z q_1]$
3. $q_0 Z q_0 \to 1[q_0 B q_0][q_0 B q_0] \mid 1[q_0 B q_1][q_1 B q_0]$
4. $q_0 Z q_1 \to 1[q_0 B q_0][q_0 B q_1] \mid 1[q_0 B q_1][q_1 B q_1]$
4. $q_0 Z q_0 \to 1[q_0 A q_0][q_0 A q_0] \mid 1[q_0 A q_1][q_1 A q_0]$
$q_0 Z q_0 \to 1[q_0 A q_1][q_1 A q_1]$

855
5162
TA

Bloom's Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	10	13	24	13	7	21	22	10	-	-

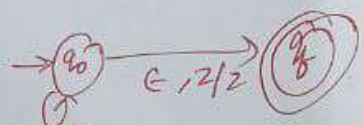
6.a) G_1: $S_1 \to AB$, $A \to aAb \mid \epsilon$, $B \to cC \mid \epsilon$
G_2: $S_2 \to CD$, $C \to aA \mid \epsilon$, $D \to bDc \mid \epsilon$

Course Outcomes:

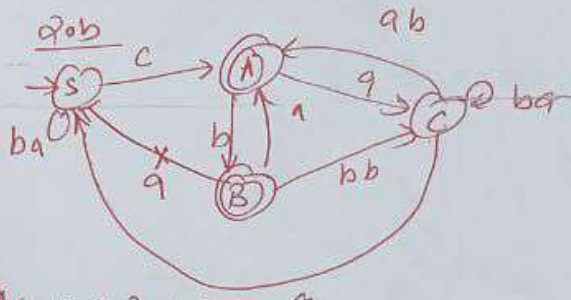
1. Understand the fundamental concepts of theory of computations
2. Analyze the tools of finite automata to various fields of computer science.
3. Design solution model for complex problems, using the appropriate skills of automata theory for better results.
4. Apply automata skills in situations that describe computation effectively and efficiently.

****

2.a



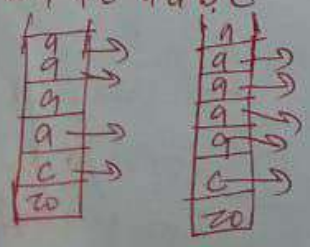
a, z / az
b, z / bz
c, z / cz
a, b / ε
b, a / ε
a, c / ε
a, a / aa
b, b / bb
b, c / bc
c, a / ε
c, b / cb
c, c / cc



3.b $(q_0, w, z_0) \vdash^* (q_f, \epsilon, \alpha)$
$(q_0, w, z_0) \vdash^* (q_f, \epsilon, \epsilon)$

4.a) Eliminating ε-production
$S \to abAB \mid abA \mid abB \mid ab$
$A \to bAB \mid bA \mid bB \mid b$
$B \to BAa \mid Ba \mid Aa \mid a \mid A$

Unit-production Elimination
$S \to abAB \mid abA \mid abB \mid ab$
$A \to bAB \mid bA \mid bB \mid b$
$B \to BAa \mid Ba \mid Aa \mid a \mid bAB \mid bA \mid bB \mid b$

Remove useless symbol No useless
$S \to X_4B \mid X_3A \mid X_3B \mid X_aX_b$
$A \to X_5B \mid X_6A \mid X_6B \mid b$
$B \to BX_6 \mid BX_a \mid AX_a \mid a \mid X_5B \mid X_6A \mid X_6B \mid b$
$X_a \to a$, $X_b \to b$, $X_3 \to X_aX_b$, $X_4 \to X_3A$, $X_5 \to X_6A$
$X_6 \to AX$

cgacaabc